Peer-to-Peer Architecture & BitTorrent Case Study

Coverage: P2P Scalability Time Models, BitTorrent Swarms, Trackers & DHT, Chunks, Rarest-First, Tit-for-Tat Choking

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1. Scalability Models

- Client-Server Time: D_CS >= max(N*F / u_s, F / d_min). Linear growth with N.
- P2P Time: D_P2P >= max(F / u_s, F / d_min, N*F / (u_s + sum(u_i))). Sub-linear / bounded growth.

2. BitTorrent Mechanics

- Swarm: Group of active peers. Chunks: 256KB - 2MB pieces. Rarest-First chunk selection.
- Tit-for-Tat (TFT): Top 4 contributing uploaders unchoked every 10s; Optimistic unchoking every 30s.

■ KTU University High-Yield Exam Questions

- Q: Derive the theoretical file distribution time for Client-Server and P2P architectures. (8 Marks)